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Orbital Gaze, Earthly Wounds: Reading Samantha Harvey's *Orbital* as Planetary Fiction

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ABSTRACT
This article examines Samantha Harvey's *Orbital* as a representative work of planetary fiction. From the orbital vantage point of International Space Station (ISS), the novel presents Earth as a border-less planet and promotes an eco-cosmopolitan ethic beyond nations and species. By portraying the consequences of nuclear catastrophes and human-induced climate change, the novel also zooms in on the personal experiences of ecological refugees. Drawing on the concept of "planetarity" first proposed by Gayatri Chakravorty Spivak, and later developed by Dipesh Chakrabarty, Timothy Clark, Amy Elias, Christian Moraru, among others, we observe that *Orbital*, through a defamiliarized narrative that views the Earth as a distant, living, and integrated planet, challenges anthropocentric thinking and reimagines human-nonhuman relations. *Orbital* urges us to reconsider the sense of planet in the contexts of the Anthropocene and Capitalocene, and retrace the metabolic rift on Earth. We argue that this orbital vantage point functions as a unique narrative device that renders visible the planetary scale of ecological degradation, and thus demonstrates planetary fiction's capacity to represent planetarity, envisioning our collective future and new forms of responsibility on the increasingly uninhabitable planet Earth.

Introduction

Samantha Harvey's Booker Prize winning novel *Orbital* portrays the collective life of six people on a space station circling the Earth.[1] The novel adopts a planetary perspective to view the Earth, with a synthesized narration of each of these six people's life stories. Since its publication, the novel has received wide recognition, and has been examined through multiple critical lenses. Maytham Abbas employs Merleau-Ponty's theories of the lived body to explore the astronauts' transformed embodiment, altered consciousness, and the emergence of the overview effect – the cognitive and emotional shift resulting from viewing Earth from space (57). Tyne Sumner argues that *Orbital* functions as a lyrical and introspective narrative experiment in socially extended cognition, depicting isolated astronauts who gradually merge into a "group mind" or "single collective consciousness" across their nine-month voyage, thereby testing the limits of individual perception and the embodiment in interstellar space (122). Situated within the spatial turn in literary theory, Philip Howell applies Object-Oriented Ontology (OOO) to frame an object-orientated literary geography, analyzing *Orbital* through the theme of disconnection stemming from the necessary withdrawal of real objects from human perception (121). Notably, several key studies on Harvey's *Orbital* interrogate ecological, planetary, and Anthropocenic concerns in the novel through diverse theoretical lenses. Claire Cazajous-Augé conceptualizes the "Anthropocene cosmic sublime" rooted in the astronauts' embodied encounters, arguing that this perspective shifts from traditional terrestrial aesthetics to a renewed

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ethical consciousness of the cosmos (97). Md Samiul Azim argues that the novel utilizes an experimental orbital spatiality to reveal the fragility of Earth and the existential insignificance of humanity, fostering a planetary ethic by merging ecological crisis with existential reflection (120). Constanza Mondo analyzes how the representation of "object Earth" and the super-typhoon in the novel reflect the distinction between global and planetary scales (129). These studies provide vital insights into the aesthetic and phenomenological dimensions of the planetary consciousness as depicted in *Orbital*. However, they rarely articulate how the logic of capitalist modes of production and accumulation functions as the systemic engine for the ecological degradation on the planet Earth, thus isolating the planetary vision from the material socio-political consequences and localized human suffering in the novel. This article extends these discussions by reading *Orbital* as a quintessential example of planetary fiction. Drawing on the concept of "planetarity" inaugurated by Gayatri Chakravorty Spivak and subsequently developed by Dipesh Chakrabarty, Timothy Clark, and Ursula K. Heise, among others, this article approaches *Orbital* from a planetary perspective. We argue that the orbital vantage point functions as a rigorous narrative device that bridges the cognitive chasm between the planetary scale and the material realities of ecological degradation on Earth. By rendering visible the metabolic rift in the Capitalocene and the lived crises of ecological refugees in the Anthropocene, Harvey's narrative moves beyond the aesthetic awe to an eco-cosmopolitan ethic of planetarity——one that fosters a collective, transborder responsibility for an increasingly uninhabitable Earth.

Planetarity and Its Literary Representations

The planetary turn is a significant transformation within contemporary literature and theory arising in the twenty-first century. It is viewed as a critical paradigm shift, appearing after previous critical movements such as the postcolonial turn, postmodern turn, and global turn (Elias & Moraru xi). The concept of "planetarity" provides a crucial critical framework for this article to understand the complex challenges of the Anthropocene, such as climate change. This concept encourages us to rethink humanity's place in the history of the Earth and move beyond traditional dichotomies that separate humans from nature.

The term "planetarity" was first used by Spivak in her address to a Swiss organization in 1997. Spivak first articulates "planetarity" as an ethical and epistemological alternative to the homogenizing logic of globalization (450). Masao Miyoshi's essay "Turn to the Planet: Literature, Diversity, and Totality" advances this perspective by calling for a planetary consciousness capable of addressing ecological and cultural interdependence on a global scale (295). The framework is later consolidated in Elias and Moraru's edited volume *The Planetary Turn: Relationality and Geoaesthetics in the Twenty-First Century*, which synthesizes the study of literature through the lens of planetary relationality and geoaesthetics. Elias and Moraru stress the ethical engagement inherent in this planetary perspective, which also serves as a "multicentric and pluralizing" model for understanding our interconnected world (xxiii). Chakrabarty underscores the urgency of the planetary perspective, noting that the current "planetary crisis of climate change or global warming" elicits a range of human responses and profoundly shapes our "sense of the now" (23). This crisis, according to Chakrabarty, reveals humanity's emergence as a "geological agent" capable of impacting the planet on a scale comparable to forces that cause mass extinctions, thereby collapsing "the distinction between human and natural histories" (31). Timothy Clark further compels us to "think on a planetary scale"(21) to grasp the true scope of global environmental issues, given that these issues "cannot immediately be seen or localized" (175). Similarly, Timothy Morton describes planetary problems like global warming as "hyperobjects," one of whose elusive features is their non-locality (38). These ecological issues cannot be observed in any single place but are registered statistically through their combined, distributed effects.

Literary imagination of the planetary can be traced throughout world literary history. Ancient world literature embodies early forms of planetary thinking, depicting humanity's expanding awareness of its relation to the Earth and its limits (Puchner 43). *The Epic of Gilgamesh* and Homer's *Odyssey* imagine the tension between civilization and wilderness, settlement, and mobility, that continues to

shape planetary narratives (59). Wai Chee Dimock interprets *The Epic of Gilgamesh* as a narrative demonstrating a "less-than-human baseline" where human and nonhuman mortality intersects (130). Dimock argues that the ancient epic functions as a mutable, recycled genre because it is co-articulated with processes of decomposition and recomposition that sustain a planetary ecology, with science fiction serving as its modern mutation (131). By the early modern period, works like Thomas More's *Utopia* anticipate later planetary frameworks by imagining global-scale communities. In the late nineteenth and early twentieth centuries, early forms of planetary imagination appeared in planetary romances and planetary adventures, as seen in H. G. Wells's *The War of the Worlds* and Edgar Rice Burroughs's *A Princess of Mars*—which dramatize encounters with other worlds and the limits of human knowledge. Wells, in particular, explores the political tension between the conceptually ideal "world state" and the materially constrained "planet," as evidenced in *The War of the Worlds*, where the Earth's biological environment ultimately defeats the Martian "world" (Lee 241).

While these early works project planetary otherness through exotic settings and interplanetary travel, contemporary planetary fiction reconfigures this tradition under the pressures of the Anthropocene, shifting its focus from adventure to ethical coexistence, deep time, and planetary-scale thinking. During the mid-to-late twentieth century, canonical science fiction writers such as Frank Herbert, Ursula K. Le Guin, and Kim Stanley Robinson expanded the imaginative and ethical horizons of the planetary mode. Herbert's *Dune* explores the interdependence of environment and power on Arrakis; Le Guin's *The Left Hand of Darkness* uses the planet Gethen to question gender and culture; and Robinson's *Mars Trilogy* envisions planetary colonization as political ecology. In the twenty-first century, planetary fiction directly engages the climate crisis through formal innovation. Robinson's *The Ministry for the Future*, for example, develops a "planetary polyphony" that integrates geophysical actants and the Earth alongside human, collective, and technological voices in a strategy of cross-scale recursion (Patoine 146). The antipodean perspective found in the works of Australian writers such as Tim Winton, Gail Jones, and Alexis Wright functions as a geographically specific intellectual legacy that counters abstract universalism, which embodies planetary thinking by grounding it in local, material, and dynamic relations (Giles 147). To date, planetary fiction functions as a critical mode that pushes literary representation beyond national or global confines toward an understanding of human and nonhuman coexistence and ethical responsibility on a complex and ever-changing planet.

Against this backdrop, this article takes *Orbital* as a crucial case study of planetary fiction, one that crystallizes the entanglement of utopian community, planetary ecological degradation, displacement of refugees, and global inequality, demonstrating literature's capacity to theorize planetarity in ways that complement and complicate the existing critical discourse.

Orbiting Planet Earth: Utopian Community Beyond Borders

Harvey's *Orbital* offers a meditation on the Earth from a planetary perspective. Viewed from space, the outlines of political borders disappear, and planet Earth emerges as a "rolling indivisible globe" with "no possibility of separation" (Harvey, *Orbital* 72). This view from the ISS (International Space Station) not only reveals the interconnectedness of the Earth's systems but also destabilizes the divisions that separate humans from other forms of life. In contemporary critical discourse, the term "orbit" emerges as an "ambivalent signifier," a complex and elusive concept encompassing the known and unknown, the visible and the imperceptible (Leonard 28). By erasing both national boundaries and anthropocentric hierarchies, Harvey reimagines community as a network of shared existence that cuts across nations and species alike. In place of territorial divisions and narrow national loyalties, the novel advances a planetary ethic grounded in ecological awareness, interdependence, and responsibility.

In the novel, the ISS becomes a miniature symbol of global community. The story happens in only twenty-four hours, following six astronauts and cosmonauts.[2]Through their daily life in the orbit, Harvey brings out questions of how we value life on the Earth and the possibility of human unity. The

ISS itself has both historical and practical meaning. The construction of the ISS started in 1998 as a joint project of the United States, Russia, Europe, Japan, and Canada. Its main purpose is to provide a platform for scientific experiments in microgravity,[3] which are important for future long-term human space travel. ISS symbolizes international cooperation, not only in science but also in diplomacy. It was once seen as a physical sign of globalization and prosperity, even praised as "the most wide-ranging international cooperative space program in history" (Castaño 672). ISS was designed to last about fifteen years, but it has already worked for nearly twenty-eight years. However, when the novel *Orbital* was published, ISS was scheduled to retire in 2030.

In the current world of increasing geopolitical conflicts, Harvey sees this "space-age relic" as especially meaningful. In her essay "Space Pastoral: Finding a New Literary Genre in the Slow Death of the International Space Station," she describes ISS as a "new-millennium shepherd," quietly watching humanity and the universe. To her, the station gives a special impetus for writing, connecting outer space with time and memory. ISS carries "a lost era, an era that was once our golden future." *Orbital* can therefore be read as a retro-futurist work, which frames the future as a site of nostalgia for the past. It is not typical science fiction, but closer to "sci-fact," because the narrative is based on scientific details and the real experience of orbital life. As Kiran Sitole notes, the novel paints "the practical aspects of life in space station" by seamlessly integrating "real scientific phenomena" into its storytelling (295–96). Within this retro-futurist frame, Harvey expresses her hope for a shared human community, using the view from space to rethink how humanity lives together on the Earth and relates to the planet itself.

From space, the Earth presents itself as a continuously turning, indivisible sphere – dynamic and visually striking. For astronauts in the low orbit, this vision becomes multidimensional and affective, linking perception with planetary scale. This vision challenges the permanence of national divisions. At night, city lights reveal "only one man-made border in the whole of the world" – "a long trail of lights between Pakistan and India." In daylight, however, that line vanishes into "the humanless simplicity of land and sea" (72). Imagined instead as "endlessly connected, an epic poem of flowing verses," the image gestures toward a different vision of the globe (124). Such switch of viewpoint demonstrates the artificiality of geopolitical boundaries and evokes the possibility of a world beyond them.

Resisted by ground control, which regards borderless thinking as contrary to national interests, the astronauts in the station experience the absurdities of geopolitical rivalry through mundane details – segregated toilets marked "Russian Cosmonauts Only" and "American, European and Japanese Astronauts Only." Yet daily survival in a closed environment compels them to embrace a different identity: that of a "nationless, borderless outpost" (63). Sharing recycled air and water makes division impossible; as they note, "we are one." In this sense, their experience embodies a "deterritorialized" perspective that exposes the fictitious nature of political borders and affirms planetary ecological unity (Deleuze & Guattari xvii). This outlook also resonates with Heise's concept of "eco-cosmopolitanism," which stresses that "the Earth's inhabitants, regardless of their national and cultural differences, are bound together by a global ecosystem whose functioning transcends humanmade borders" (25). The concept of a new "planetary flag" is introduced at the end of the novel, as a potential symbol for a unified, interstellar society:

> Maybe we're the new dinosaurs and need to watch out. But then maybe against all the odds we'll migrate to Mars where we'll start a colony of gentle preservers, people who'll want to keep the red planet red, we'll devise a planetary flag because that's a thing we lacked on earth and we've come to wonder if that's why it all fell apart, and we'll look back at the faint dot of blue that is our old convalescing earth and we'll say, Do you remember? Have you heard the tales? Maybe there's another parent-planet—earth was our mother and Mars, or somewhere, will be our father. We are not such orphans-in-waiting after all. (Harvey, *Orbital* 195)

The "planetary flag" is conceived as a symbol of unity and a corrective measure for perceived failures on the Earth, where the lack of such a unifying sign is hypothesized to have contributed to fragmentation and conflicts. In this sense, this unprecedented creation embodies both a utopian aspiration for

cohesion and a retrospective critique of terrestrial history, offering a reimagined foundation for collective belonging beyond the Earth. At the same time, the astronauts' extreme isolation forces them to reconstruct community in intimate human terms. To endure loneliness within the "brutal, inhuman, overwhelming" conditions of space, they reimagine themselves as parts of a single organism, where "they're all there is" and must serve as "everything to each other for this short stretch of time" (17–18). Each member is figured as a vital organ – Anton as the heart, Pietro as the mind, Roman as the hands, Shaun as the soul, Chie as the conscience, and Nell as the breath. Through this metaphor, humanity becomes a "near and graspable thing," transcending "confounding difference and distance" (18). They gradually realize a profound desire to remain in the orbit, recognizing that the fragile family they have formed there is the only one capable of fully sharing and comprehending their singular experience. This community, forged not through kinship or nationality but through the extraordinary conditions of space, redefines belonging itself. Suspended between the Earth and the infinite beyond, they come to understand that intimacy and solidarity arise less from inherited bonds than from the shared vulnerability and interdependence that orbital life demands.

Beyond the portrayal of the astronauts' heartfelt connections unbounded by nationality or identity, the novel also illuminates their empathy and their enduring ties to the living world below. Even removed from the Earth, they recall with longing the presence of animals – a "curious dog, a startled rabbit, a leaping sudden deer," along with "gibbons," "horse," "jay," "spider," "pike," "shrew," "hare," and "scarab beetle" – as well as the memory of trees (11). Such recollections signal an irreducible connection to a multispecies environment. From the ISS, they are also able to witness meteorological events and patterns of animal migration, where images from the orbit appear to "confirm what the eddying birds and running goats already seem to know," such as the formation of a typhoon (37). This juxtaposition suggests a continuity from the scientific observation by humans to the intuitive knowledge of non-human species, pointing to a shared sensitivity to planetary processes. This kind of "species thinking," resonates with Chakrabarty:

> The consequences make sense only if we think of humans as a form of life and look on human history as part of the history of life on this planet. For ultimately what the warming of the planet threatens is not the geological planet itself but the very conditions, both biological and geological, on which the survival of human species as well as of other forms of life depends. The widely acknowledged threat that the present crisis of biodiversity may indeed balloon into a sixth great extinction of species in the history of the planet constitutes an event horizon for several mainstream narratives of planetary climate change. (36)

Chakrabarty calls for a biocentric reframing of human history within the broader continuum of life on Earth. He challenges the anthropocentric tendency to isolate human affairs from planetary processes by asserting that the meaning of climate consequences becomes intelligible only when humans are understood as one among many life forms. The argument shifts attention from the geological planet – the material substrate that will endure – to the conditions of habitability that sustain life. Thus, the encounters with non-human creatures in the novel are not incidental but can provoke deep reflections on coexistence.

The novel draws on Velázquez's *Las Meninas* as a symbolic lens,[4] proposing that the painting's true focus is the dog – an animal free from vanity precisely because it does not seek to be seen, standing in contrast to the self-conscious human figures. By facing the reality of human extraneousness and loosening the hold of ego, the novel suggests that peace becomes possible – not through dominance, but through the cultivation of a shared "global species consciousness," which is a critique of humanity's species arrogance (Clark 175). The astronauts, reflecting on their own situation, arrive at a similar awareness: they describe themselves as "lab rats" and "specimens," reduced to "data" and serving only as a "means and not an end" for future interplanetary ventures, highlighting their growing sense of human "smallness and briefness" (Harvey, *Orbital* 94). This is in accordance with Donna Haraway's notion of "sympoiesis" (co-making or becoming-with, 58) and her proposal of the "Chthulucene" (99), both calling for rejecting human exceptionalism in favor of "multispecies worlding on a wounded terra"(105). This kind of respect also extends to the Earth itself, imagined as a breathing organism, "an

animal unto itself" (Harvey, *Orbital* 72). Such imagery frames the planet as alive and autonomous, challenging any notion of it as merely a stockpile of resources. This perspective situates human activities within a broader, dynamic system, emphasizing that sustainable survival depends not on domination but on attentiveness, care, and a recognition of Earth as an active participant in the ongoing web of life.

As discussed above, *Orbital* uses the astronauts' vantage point to dissolve the boundaries that humans draw between themselves and other forms of life. The novel presents a fragile, interdependent network – in which human survival is intimately connected not only to countless other species but also to the Earth as a living, dynamic whole. By highlighting these reciprocal relationships, the narrative emphasizes that well-being and continuity depend on awareness, empathy, and ethical engagement with the broader ecological system, suggesting a vision of existence in which humans are inseparable from the life-supporting processes of the planet.

Tracing Capitalocene: Planetary Scale of Matabolic Rift

Harvey's *Orbital* presents a haunting view of the Earth's ecological collapse, seen through the unprecedented perspective of astronauts orbiting above. This literal and metaphorical vantage point acts as a key narrative strategy, revealing the planetary scale of environmental damage under late capitalism. The novel's fragmented scenes – ocean pollution, vanishing rainforests, soil erosion, and extreme weather events – combine to form a strong critique of the Capitalocene. Jason Moore argues that Capitalocene is a more politically sharp and historically accurate term than Anthropocene, which he criticizes for too easily blaming ecological disasters on an "undifferentiated humanity" (the "Anthropos") (108). Instead, Capitalocene focuses on capitalism's structural forces, driven by relentless accumulation, as the main cause of planetary change. Haraway also supports this view, stating: "Species Man did not shape the conditions for the Third Carbon Age or the Nuclear Age," emphasizing that "capitalism" or "Capitalocene" provides a better framework for understanding such major shifts (47). This perspective helps us see the ecological crises as the result of specific economic, social, and power structures of our world. Harvey's *Orbital* shows how the logic of capital – based on endless growth and ignoring ecological costs – transforms the planet: forests become cattle ranches; rivers turn into static scars; and the atmosphere fuels monstrous storms. However, as Clark points out, these global events like deforestation, climate change, and gradual ecological damage are hard to see at the "normal human scale" (x). This gap between what we experience and the planetary reality often makes ecological destruction seem distant or theoretical. *Orbital*, from its orbital viewpoint, breaks down this distance. The narrative forces us to recognize how deeply nature has been altered by human actions, especially those driven by wealthier nations and commercial interests.

Within the context of the capital world, the degradation of freshwater and marine systems has become an important indicator of ecological crises. By depicting polluted oceans, contaminated rivers, and shrinking lakes, this novel reveals how capitalism is eroding the foundations of life on Earth. From their orbital vantage point, the astronauts see "swirling neon or red algal bloom in the polluted, warming, overfished" waters in the Atlantic, which reflects both toxic accumulation and over-exploitation (Harvey, *Orbital* 74). This relentless process is what Rob Nixon calls "slow violence," a kind of environmental harm that works invisibly across generations (6). Capitalism's pursuit of endless profit externalizes its costs to the oceans, turning them into dumping grounds as well as resource frontiers. From space, this logic becomes visible in the sea itself, which has been transformed into evidence of systemic crisis. The process of capitalization also produces serious industrial pollution. "Every burning oil spill" demonstrates the persistence of industrial negligence, while freshwater systems show multiple signs of contamination. Examples include the "discolouration of a Mexican reservoir which signals the invasion of water hyacinths feeding on untreated sewage," the "distorted flood-bulged river in Sudan or Pakistan or Bangladesh or North Dakota," and the "prolonged pinking of evaporated lakes"(Harvey, *Orbital* 74). In addition, the ecological footprint of mining is clear in the "expanding green-blue

geometries of evaporation ponds where lithium is mined from the brine," which shows how resource extraction reshapes landscapes and damages local environments (74). Together, these scenes present a breakdown of hydrological cycles and highlight the failures of environmental governance under capitalism. Carolyn Merchant, the pioneering ecofeminist scholar, has documented the criticisms of mining in *The Death of Nature*, describing mining as a central example of humanity's destructive interaction with the Earth. She notes that mining not only pollutes water but also damages the subterranean environment, so the "living earth" is seen as being "defiled by pollution of 'its purest streams' and by the mining of its entrails for metals" (Merchant 129). The novel links capitalist exploitation to both visible and hidden ecological damage, emphasizing how economic activities of humans reshape natural systems and reveal the deep interconnections between social, economic, and environmental processes.

On land, the destruction continues. The astronauts see "Gran Chaco's brown seepage of cattle ranch where once was rainforest," and the "vanishing mangrove forest in Mumbai"(Harvey, *Orbital* 74–75). These important ecosystems are severely damaged, and this loss is closely connected with livestock industry and forestry engineering. The astronauts also see "a thousand-mile patchwork of orange dots where the rainforest burns" in South America – "crossing a continent on fire"(126). This indicates that deforestation and wildfire are happening on a very large scale, and they are not separate events. They are connected with land exploitation, cattle ranching, and global commodity demands. The French historians Fabien Locher and Jean-Baptiste Fressoz point out that "the climatic impact of deforestation could be considered on a planetary scale," causing serious problems such as droughts, floods, bad seasons, and abnormal growth of polar ice caps (585). The increasing frequency of wildfires is not merely a natural phenomenon; rather, it is a symptom of profound ecological damage – an attritional and often invisible form of environmental harm that steadily erodes ecosystems across time and space. The rainforest's water cycle, which used to protect it from fire, is partly broken. When the rainforest burns, it stops being a carbon sink and instead releases a large amount of carbon dioxide. This carbon release accelerates global warming and creates a vicious circle, because this causes higher temperature, which in turn induces more fires. The Earth's circular systems – growth and decomposition, rainfall and evaporation, alive with the cycling of air currents, are interconnected systems that cannot be disturbed. Therefore, the destruction of forests is both a consequence and a cause of environmental crises, and it reflects the ecological cost of capitalist expansion that prioritizes profit and production over nature.

Industrialization, modernization, and urbanization have reshaped land on a planetary scale. In the novel, humans are not seen in the daytime – humanless – but at night humans become visible: "mankind is the light of cities and the illuminated filament of roads"(14). Technological power of human beings dominates the environment, but also leaves visible marks of ecological damage. In addition, "altered contour of a coastline where sea is reclaimed metre by painstaking metre and turned into land to house more and more people" (75). There are "hundreds of acres of greenhouses which make the entire southern tip of Spain reflective in the sun"(75). This "plastic sea" of greenhouses in Almería, southern Spain, manifests the negative effects of industrial agriculture. The vast area of greenhouses demonstrates human reshaping of nature. These scenes indicate that industrialization and modernization, driven by capitalist logic, disrupt ecosystems and compromise environmental integrity.

Anthony Giddens links industrialization, modernization, and capitalism directly to environmental degradation, arguing that growth which damages the biosphere is "uneconomic" and that orthodox economics reduces both human life and nature to mere "factors of production alongside other commodities" (52–53). Capitalist activities – industrial farming, deforestation, and urban expansion – emerge as key drivers of ecological destruction. This planetary reconfiguration of land is driven by the forces of capitalist expansion and technological modernization that have intensified human intervention in the Earth system. In the novel, the astronauts, gazing upon the wounded surface of the Earth from orbit, come to recognize the profound consequences of human-driven transformation:

> It's utterly manifest in every detail of the view, just as the sculpting force of gravity has made a sphere of the planet and pushed and pulled the tides which shape the coasts, so has politics sculpted and shaped and left evidence of itself everywhere. They come to see the politics of want. The politics of growing and getting, a billion extrapolations of the urge for more, that's what they begin to see when they look down. They don't even need to look down since they, too, are part of those extrapolations, they more than anyone—on their rocket whose boosters at lift-off burn the fuel of a million cars. The planet is shaped by the sheer amazing force of human want, which has changed everything, the forests, the poles, the reservoirs, the glaciers, the rivers, the seas, the mountains, the coastlines, the skies, a planet contoured and landscaped by want. (Harvey, *Orbital* 75)

The ecological degradation unveils the logic of sacrifice zones – places beyond repair through destructive extraction and irreversible pollution. It is inherent to a capitalist perspective that reduces the environment to "cheap nature" (Farrier 52; Moore 79). The astronauts, enmeshed in this logic, recognize their own complicity when their boosters "burn the fuel of a million cars," emitting a huge amount of greenhouse gases (Harvey, *Orbital* 75). The novel illuminates how the capitalist drive for endless growth and exploration transforms both the material cycles of nature and the geographies of the planet, binding human desire to landscapes of loss and ecological precarity.

Harvey's orbital narrative perspective achieves a strong sense of defamiliarization. It forces readers to recognize the planet as one interconnected system, deeply wounded by the extractivist logic of capitalism. The burning Amazon, the drying rivers in America, the intensifying typhoon, the algae-choked reservoirs, the lithium ponds, and the space debris are not isolated events; they are interconnected symptoms of this planetary metabolism in crisis. Such representations exemplify what Karl Marx, as articulated by John Bellamy Foster, calls the "metabolic rift": a systemic rupture in the material cycles between society and nature caused by capitalist production, an irreparable break in the interdependent processes of social metabolism (Foster 163). *Orbital* visualizes the Capitalocene as a condition of widespread ecological degradation shaped by the law of extraction and accumulation, revealing capitalism as a geological force not through abstraction but through the visceral, interconnected scars it leaves on the living Earth and its surrounding space.

Planetary Exile in the Anthropocene: Humanity without Refuge

Donna Haraway, building on Anna Tsing's work, describes the Anthropocene as a pivotal era defined by the loss of refugia. Once these shelters disappear, the planet becomes "full of refugees, human and not, without refuge" (100). According to Haraway, the central "outrage" of the Anthropocene is this destruction of the spaces and times that make survival possible and she suggests that humanity's essential task is to "make the Anthropocene as short/thin as possible" (100). This conceptualization foregrounds the sense of exile in the Anthropocene: not simply migration across borders, but the planetary-scale erosion of safety, security, and habitability. As Moraru suggests, planetary fiction should also "telescope" the world in a way that both condenses and enlarges it, bringing it closer and spreading it out (236). While framing macro scale crises such as global warming, *Orbital* simultaneously zooms in on the micro level problems – the suffering and trauma of ecological refugees. In this way, the novel telescopes the planetary and the personal, condensing planetary crises into singular, embodied stories.

The concept of "refugee" has evolved in the context of environmental challenges. The term "environmental refugee" first emerged during the 1970s and 1980s in discussions surrounding the environment, migration, and population. In *Environmental Refugees*, Essam El Hinnawi defines "environmental refugees" as individuals who are forced to leave their home due to significant environmental disruptions, whether natural or man-made, that threaten their survival or severely affect their quality of life (4). Since 2008, an average of 26.4 million people have been forcibly displaced annually by catastrophic environmental events such as hurricanes, tsunamis, earthquakes, and other extreme weather occurrences, making "climate refugee" an increasingly prominent term in public and political discourse (Apap 1). In *Environmental Justice and the Rights of Ecological Refugees*, Laura Westra adopts the term "ecological refugees" as a comprehensive

definition covering individuals, families, and populations displaced by severe environmental disruption. The term "ecological refugees" deliberately subsumes "environmental refugees" and "climate refugees," while also extending to "those fleeing industrial and chemical hazards" (xviii). These terms underscore the growing severity of refugee issues in the context of various ecological and environmental disasters.

In *Orbital*, Harvey explores the long-term impacts of both historical and contemporary human-induced crises, depicting a world increasingly inhabited by people who are displaced and without refuge. The novel captures the devastating effects of such events, moving from the personal trauma caused by the nuclear aftermath to the widespread displacement brought on by climate-related disasters. Through this, Harvey frames exile not as an isolated or rare event but as an ongoing and cumulative condition of modern life.

Chie's family history in the novel offers a personal account of nuclear destruction, particularly the atomic bombing of Nagasaki in 1945. Her mother's survival is depicted as a miracle, as she narrowly escaped death due to a series of fortunate circumstances – her grandfather's illness and being at home with the baby on the day the bomb fell (Harvey, *Orbital* 59). In contrast, her grandmother was killed in the Nagasaki munitions factory, leaving "no remains" and highlighting the extreme and permanent destruction that such "moments of glory" can cause (59, 61). This personal history of survival, shaped by a human-made catastrophe, leaves a lasting impact on Chie. She imagines that her mother silently conveyed a sense of fear when looking at the sky on Moon Landing Day, offering a quiet warning to "be afraid, my child, at what humans can do" and to "never forget the price humanity pays for its moments of glory, because humanity doesn't know when to stop" (61).

The impacts of nuclear accidents and explosions, such as the Fukushima Daiichi disaster, can lead to severe consequences, including mass displacement and long-term challenges for affected populations. Areas around such sites become "difficult-to-return zones" due to dangerous radiation levels (Tanaka 429). In addition to the immediate physical threats, exposure to radiation can cause significant psychological stress due to the uncertainty of the future, especially for those who are forced to live in shelters or temporary housing and this stress may lead to "disaster-related deaths in the long term" (429). Furthermore, the lingering effects of nuclear tests can result in a state of "displacement without moving," where communities remain in their original locations, but the land and resources are so degraded that the area is no longer habitable (Nixon 19). Communities affected by nuclear testing have been referred to as "ghosted casualties" and "uninhabitants," indicating a profound loss of their traditional way of life and the inability to remain in their ancestral land (153). Despite the severity of these situations, displaced populations continue to be marginalized.

Alongside this nuclear history of displacement, *Orbital* portrays the ongoing crisis of climate refugees, specifically through the devastating impact of a super-typhoon. The storm forms over the Pacific Ocean, rapidly intensifies to a Category Five, and makes landfall across Southeast Asia, particularly in the Philippines, Indonesia, and Malaysia. Pietro, the Italian astronaut, deeply concerned for his friend, a fisherman on the "hopelessly low-lying" island of Samar in the Philippines, serves as a lens for this crisis (Harvey, *Orbital* 38). The fisherman's predicament represents the "climate refugee" experience – he cannot evacuate because he has "nothing much to lose than to keep losing something," and his "life can't be moved" (Harvey 38). His house, made of "tin, cardboard, hardboard and sticks," reflects the grim reality where frequent and powerful typhoons render rebuilding impossible (38). From the orbit, the astronauts witness the typhoon wreak havoc, turning houses into "rapids and calamitous ruin," leaving islands "misshapen by flood," and sweeping away entire communities (131). This imagery aligns with the definition of "climate refugees," who are forced to leave their homes and possessions due to catastrophic events. By showing this devastation from the astronauts' orbital viewpoint, Harvey highlights the scale of planetary exile in real time.

While terms like "environmental refugees" and "climate refugees" are used by activists and scholars to highlight their struggles, these terms remain less discussed or even unrecognized in international law, leaving these groups without due legal protection. The empirical study of Hoye et al. exhibits that the use of "environmental refugees" does not increase public awareness or urgency in countries like

UK and Netherlands (668). This lack of legal recognition and civilian concern reflects a planetary condition of *refugees denied*, where naming alone fails to secure protection.

By weaving together the legacy of nuclear catastrophes with the urgent reality of climate disasters, *Orbital* shows that exile in the Anthropocene is not confined to border crossings. It is a planetary condition, encompassing both the loss of habitable lands and the erosion of collective futures. The figure of ecological refugees – whether environmental, nuclear, or climate-related – thus becomes emblematic of a broader human condition: the shared vulnerability of living on an increasingly inhospitable planet. At the same time, the lack of legal recognition for these refugees reflects a deeper crisis of responsibility, where language and policy fail to protect those most vulnerable to ecological devastation.

Conclusion

Harvey's *Orbital* showcases how planetary fiction engages with the interconnected crises of the Anthropocene and the Capitalocene. The novel does more than depicting ecological devastation – it urges us to reconsider the ethical and affective dimensions through which such crises are understood. By setting the narrative aboard the ISS, the novel destabilizes traditional conceptions of belonging, erasing the boundaries of nations, communities, and even species. Viewed from this orbital vantage point, the Earth appears as both fragile and scarred, marked by the legacies of nuclear disaster, climate change, and capitalist exploitation. Harvey thus portrays what Donna Haraway refers to as a world "without refuge," where the spaces necessary for life are being gradually destroyed or erased (100).

Orbital does not conclude with this vision of catastrophe, however. Instead, it offers a fleeting glimpse of solidarity through the astronauts' fragile, yet borderless, community. This vision of solidarity does not reflect the typical utopia of technological mastery often found in speculative fiction. Rather, it is a more modest, yet profound, utopian impulse rooted in humility, interdependence, and care. Harvey implies that planetary crises must be understood as both a shared human condition and a deeply unequal distribution of risks and harms. This dual recognition is essential for rethinking how we approach the planet's future.

Orbital highlights the unique role that literature can play in environmental discourse. Unlike technical reports or policy documents, planetary fiction allows us to cultivate perception and affect: it makes ecological collapse visible on a planetary scale, while simultaneously making its consequences personal and relatable. *Orbital* serves as a reminder that survival on a damaged Earth cannot be reduced to technical fixes or geoengineering solutions. Instead, it requires a fundamental ethical reorientation and reimagination of community in ways that extend beyond territorial and anthropocentric boundaries. Harvey's novel exemplifies how literature can sustain an ongoing dialogue between crises and care, vulnerability and responsibility. *Orbital* illustrates that planetary fiction is not merely a mode of representation but a form of world-making. By reimagining exile, responsibility, and community from the vantage of the orbit, the novel indicates that literature remains indispensable for rethinking how we inhabit our planet.

Notes

1. Besides winning the 2024 Booker Prize, Samantha Harvey's *Orbital* was also awarded the Hawthornden Prize and shortlisted in the Ursula K. Le Guin Prize, the Orwell Prize in 2024, and the inaugural Climate Fiction Prize in 2025. See Harvey's personal website: www.samanthaharvey.co.uk/.
2. Cosmonaut is the specific term used for an astronaut employed by the Russian Federal Space Agency (or its predecessor, the Soviet space program). Space travelers have different titles depending on their training countries. Those trained by NASA (USA), ESA (Europe), or CSA (Canada) are called astronauts. Professionals from Russia are known as cosmonauts, while those from China are called taikonauts. See the following website: www.bbc.co.uk/newsround/57345234.
3. Microgravity is the condition in which people or objects appear to be weightless. See "What Is Microgravity?" *NASA*, 13 Feb. 2009, www.nasa.gov/centers-and-facilities/glenn/what-is-microgravity/.

4. Velázquez's iconic painting *Las Meninas* (*The Maids of Honour*, 1656) depicts the artist in his studio observing his models, with the Infanta Margarita at the center, surrounded by maids of honor, courtiers, and the Italian jester Nicolaso Pertusato. The painting's renown lies in its unseen models, King Philip IV and Queen Mariana, reflected in a mirror on the back wall, placing the viewer in their position and creating a complex interplay of observer and observed. See Michel Foucault, *The Order of Things: An Archaeology of the Human Sciences*, translated by Alan Sheridan, Tavistock, 1970, pp. 3–18.

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